

# BEARING WITNESS.



A REAL BEARING (SKF 129 HA) · PHOTO: WIKIMEDIA COMMONS, CC BY 4.0

# 41%

**of all motor failures are bearings.** The largest utility motor reliability study ever run: 6,312 motors.

ALBRECHT ET AL., IEEE TRANS. ENERGY CONVERSION, 1986

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**And most plants still check them once a month, by hand .**

# \$125,000

**median cost of ONE HOUR of unplanned downtime.** Two-thirds of plants eat at least one a month.

ABB, VALUE OF RELIABILITY SURVEY, 2023 · N = 3,215 PLANTS

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**A bearing can die inside the gap between two readings. The fix everyone sells is cloud, and this data is not allowed to leave the building .**

# AN ALWAYS-ON SCREENING AGENT THAT LIVES ENTIRELY ON THIS BOX.

It watches every window of vibration, explains what it sees with evidence a human can check, and files the work order.

**Nothing leaves the room.**



# PHYSICS DOES THE DIAGNOSIS. A LOCAL MODEL FILES THE PAPERWORK. A HUMAN SAYS YES.

## 01 · DETECT

Persistent change against the asset's own early baseline.

## 02 · EXPLAIN

Known machine frequencies labelled before anything is called a fault.

## 03 · CORROBORATE

Two independent spectral views must agree, at slip-aware widths.

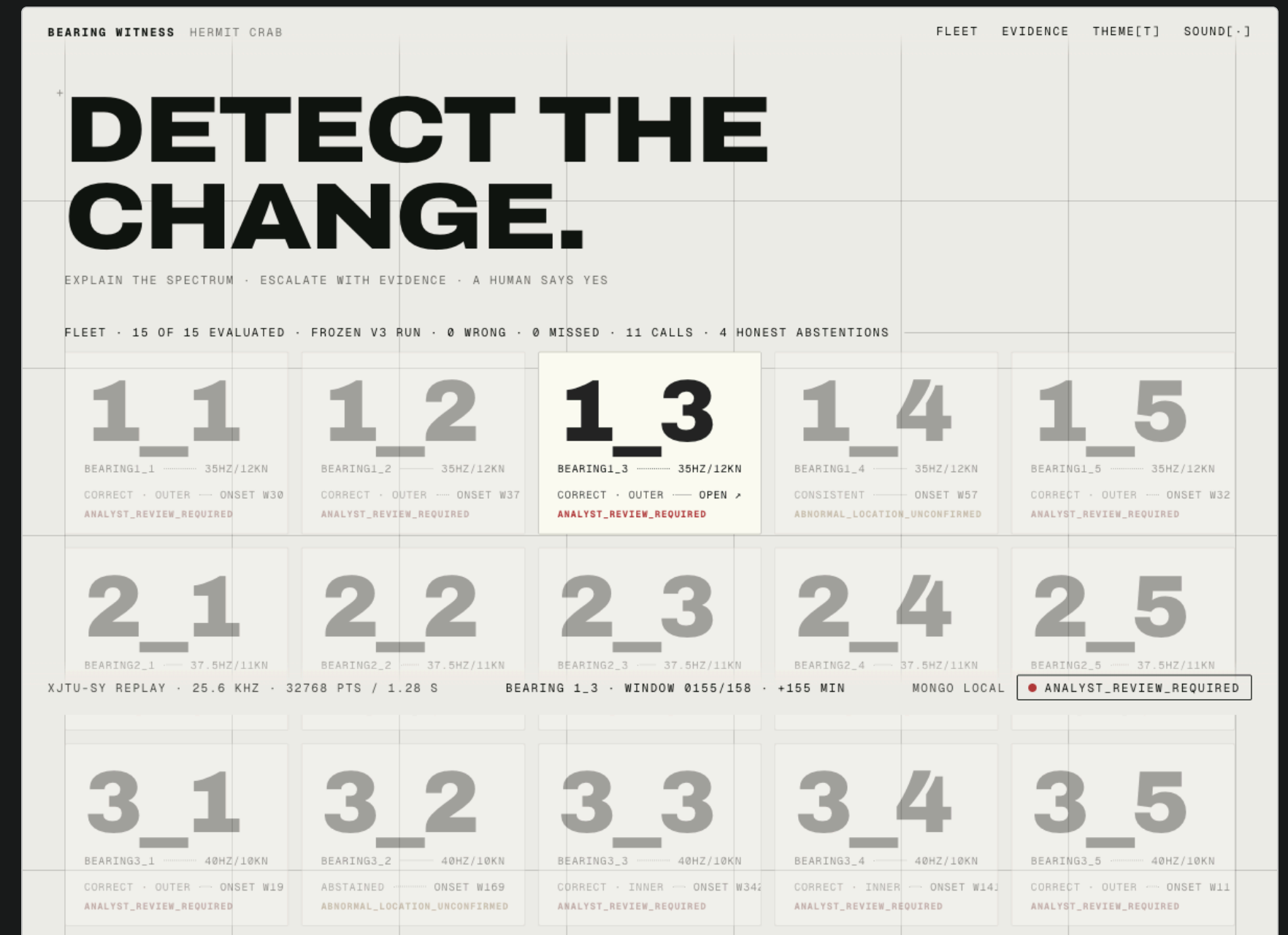
## 04 · ESCALATE

A typed work order with evidence locators. Approve, reject, or defer.

ALWAYS-ON · THE AGENT ACTS ON ITS OWN OVER TIME

# CASES LAND WHILE NOBODY CLICKS.

- Watch mode replays the bearing's life through the engine on a cadence
- Every analysis writes to MongoDB through the same gated path
- Windows it cannot prove are **refused**, and the refusal is recorded
- Zero cloud calls. The box is the whole system.



THE FLEET · 15 OF 15 EVALUATED UNDER FROZEN V3 · EVERY CARD A REAL VERDICT

BUSINESS VALUE • SOMETHING A COMPANY WOULD ACTUALLY PAY FOR

# ONE CATCH INSIDE THE MONTHLY BLIND SPOT **PAYS** **FOR THE BOX.**

## THE BUYER

Every OT-isolated plant that already refuses cloud monitoring: pharma, food, defense, chemicals, utilities. The customer who needs this most is exactly the one nothing else can serve.

AT ABB'S MEDIAN DOWNTIME RATE OF \$125,000/HOUR (2023 SURVEY, N=3,215)

## THE PLANNER'S MORNING

**Before:** a monthly route, a clipboard, a guess.

**After:** a screening queue with evidence, and the work order already in the system of record.

THE RECEIPTS · THRESHOLDS FROZEN BEFORE THE RUN · ALL 15 RUN-TO-FAILURE BEARINGS

0 / 0

WRONG CALLS / MISSED DETECTIONS

10 exact + 1 cage-consistent localizations. 4 honest abstentions, each carrying its machine reason. Inner and outer race faults both.

99 MIN

MEDIAN WARNING BEFORE END OF LIFE

42 HOURS

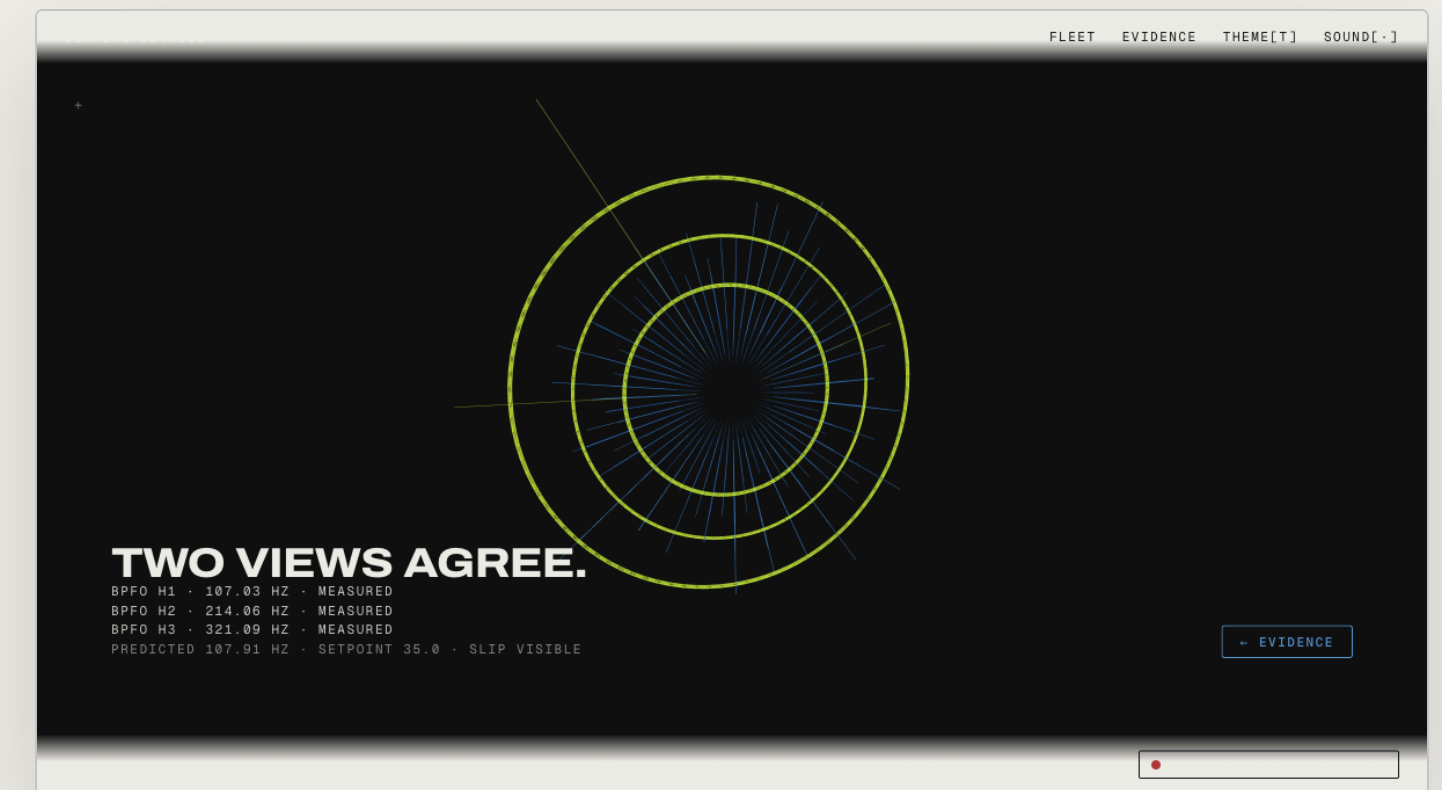
BEST CASE (BEARING3\_1, 2,519 MIN)

MEDIAN 98 EXCLUDING ONE STRUCTURAL FLOOR · WE STATE IT BOTH WAYS · FREEZE SHA IN THE COMMITTED OUTPUT



## TECHNICAL EXECUTION · IT DOESN'T BREAK

- NVIDIA NemoClaw + OpenClaw + OpenShell (deny-by-default egress, shown live)
- Local Qwen3 27B (4-bit, Ollama) on the GB10 · MongoDB Community, localhost only
- SciPy owns diagnosis: deterministic, auditable, frozen thresholds
- 150+ tests · CI green · runs from a cold clone



THE CORROBORATION VIEW · WEBGL FED BY THE REAL ENVELOPE BINS

**DETECT THE CHANGE.  
EXPLAIN THE SPECTRUM.  
ESCALATE WITH EVIDENCE.  
A HUMAN HAS TO SAY YES.**

BUILT TODAY, ON THE BOX, ON REAL RUN-TO-FAILURE DATA · IT NEVER GUESSED ONCE

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[GITHUB.COM/STEPHENSOOK/BEARING-WITNESS](https://github.com/stephensook/bearing-witness) · MIT